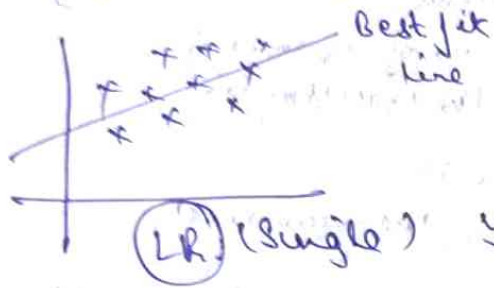


Multiple linear regression:



$x | y$ (One i/P $\rightarrow$ One o/P)

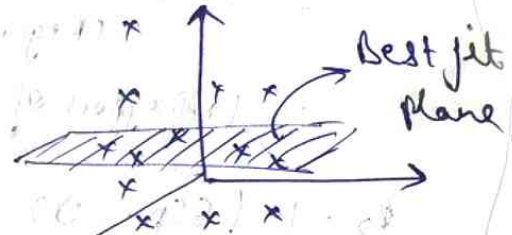
$$y = mx + b \quad (m, b) \checkmark$$

MLR:

$x_1, x_2, \dots, x_n | y$ (multiple i/P $\rightarrow$ one o/P)

Assume,

$x_1, x_2 | y$



$y = mx_1 + mx_2 + b$ (eqn of plane)

$B_1 \quad B_2 \quad B_0$

$\uparrow \quad \uparrow \quad \uparrow$

$$y = B_0 + B_1 x_1 + B_2 x_2 \quad (3D)$$

$$y = B_0 + B_1 x_1 + B_2 x_2 + B_3 x_3 \quad (4D)$$

for n dimensions,

$$y = B_0 + \sum_{i=1}^n B_i x_i$$

$B_1, B_2, B_0 \rightarrow$ weights

if B_1 is $\downarrow$, not much weightage of x_1 to calculate y .

$B_0 \rightarrow$ offset (intercept value)

Equation

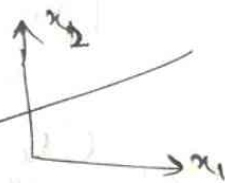
for 2D $\rightarrow$ line

3D $\rightarrow$ plane

4D, 5D ... nD $\rightarrow$ hyperplane.

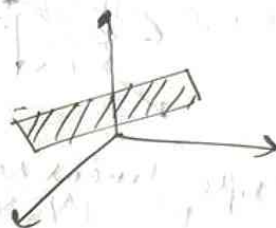
a) $ax_1 + bx_2 + c = 0$ (line)
in general,

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$



b) $ax + by + cz = 0$ (plane)
in general,

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$$



for n dimensions,

$$w_0 + \sum_{i=1}^n w_i x_i = 0$$

(Eqⁿ of hyperplane)

Hyperplane:

$$(w_1 x_1 + w_2 x_2 + \dots + w_n x_n + w_0 = 0)$$

$$w \cdot x + w_0 = 0$$

Π

$$w = [w_1, w_2, w_3, \dots, w_n]$$

$$x = [x_1, x_2, x_3, \dots, x_n]$$

$$\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0$$

$$[w_1, w_2, \dots, w_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0$$

row vector,

col vector,

if passing through origin ($w_0 = 0$)

$$w^T x = 0$$

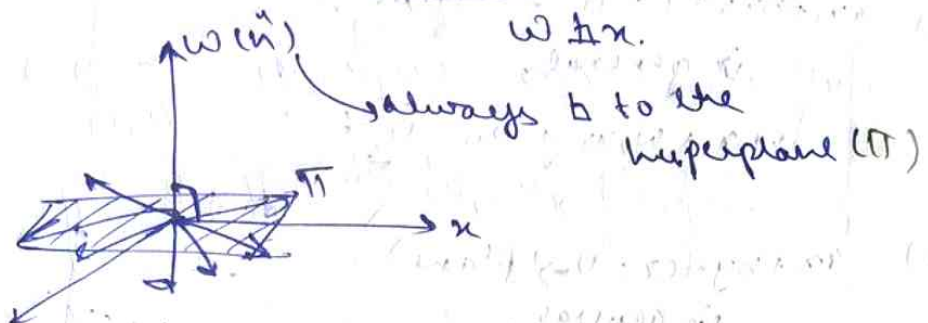
$$w_0 + w^T x = 0$$

What is w ,

$$w^T x = 0$$

$$w^T x = w \cdot x = |w||x|\cos\theta = 0$$

$$\theta = 90^\circ$$



multiple linear regression (with 3 variables)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

gpa | ce | gender | hpa

$x_1 \quad x_2 \quad x_3 \quad y \rightarrow \text{actual}$

$$(\text{predicted}) \hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

Assume 100 student data

$$\begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \vdots \\ \hat{x}_{100} \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13} \\ \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23} \\ \vdots \\ \beta_0 + \beta_1 x_{1001} + \beta_2 x_{1002} + \beta_3 x_{1003} \end{bmatrix}$$

for n rows with

$$(\beta_0, \beta_1, \dots, \beta_m) \text{ (m cols)}$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \dots & \beta_m x_{1m} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \dots & \beta_m x_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \beta_0 & \beta_1 x_{n1} & \beta_2 x_{n2} & \dots & \beta_m x_{nm} \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nm} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$$

$$\boxed{\hat{y} = X\beta}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\boxed{e = y - \hat{y}}$$

$$e = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

$$e = \begin{bmatrix} y_1 - \hat{y}_1 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

$$\boxed{E = e^T e}$$

for single R:

$$\left[E = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right]$$

(Proof?)
(NP)

$$e = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}_{n \times 1}, e^T = [y_1 - \hat{y}_1 \quad \dots \quad y_n - \hat{y}_n]_{1 \times n}$$

$$e^T e = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \dots + (y_n - \hat{y}_n)^2$$

$$[E = e^T e = \sum_{i=1}^n (y_i - \hat{y}_i)^2] \text{ (Proof)}$$

$$E = (Y - \hat{Y})^T (Y - \hat{Y})$$

$$= (Y^T - \hat{Y}^T)(Y - \hat{Y})$$

$$= Y^T Y - Y^T \hat{Y} - \hat{Y}^T Y + \hat{Y}^T \hat{Y}$$

$$\text{as } (\hat{Y} = XB)$$

$$= [X^T Y - Y^T (XB) - (XB)^T Y + (XB)^T XB]$$

$$\begin{matrix} \swarrow & \searrow \\ A & B \end{matrix}$$

$$B = A^T$$

Belowing: (Proof)

$$(A^T B = B^T A) ?$$

$$(A^T B)^T = (B^T A)^T$$

$$A B^T$$

$$A^T B = (A B)^T$$

$$[C = C^T] = ?$$

$$C = A^T B = Y^T (XB)$$

$$Y^T \rightarrow \begin{bmatrix} y_1 & y_2 & \dots & y_n \end{bmatrix}_{1 \times n}$$

$$X \rightarrow \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nm} \end{bmatrix}_{n \times (m+1)}$$

$$B \rightarrow \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_m \end{bmatrix}_{(m+1) \times 1}$$

$$Y^T \cdot X \rightarrow (1 \times n) \times (n \times (m+1))$$

$$Y^T X \cdot B \rightarrow [(1 \times (m+1)) \times ((m+1) \times 1)] = [1 \times 1]$$

So is single element,

$$(Y^T X B)^T = \text{single element}$$

$$C = C^T \text{ (proved)}$$

$$[E = Y^T Y - 2Y^T X B + (XB)^T X B]$$

for min value,

$$\frac{dE}{dB} = 0 \Rightarrow \frac{d}{dB} [Y^T X - 2Y^T X B + (XB)^T X B] = 0$$

$$= -2Y^T X + \frac{d}{dB} [(XB)^T X B] = 0$$

$$-2Y^T X + 2X^T X B^T = 0$$

$$-2Y^T X = -2X^T X B^T$$

$$B^T = (Y^T X) (X^T X)^{-1}$$

$$B = (X^T X)^{-1} X^T Y = [Y^T X (X^T X)^{-1}]^T$$

$$= ((X^T X)^{-1})^T (Y^T X)^T$$

$$= (X^T X)^{-1} (X^T Y)$$

(X) so no off of T

$$B = (X^T X)^{-1} X^T Y$$

$O(N^3)$

time

complexity

$(n+1) \times 1$

for $m=999$

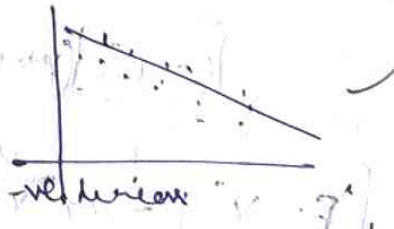
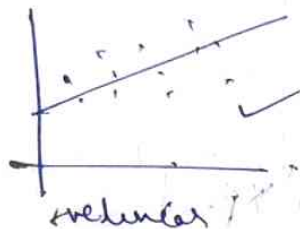
$$1000^3 = 10^9 \text{ time}$$

(Slow)

(hence gradient descent used)

Assumptions of LR:

1. Linear relationship b/w data

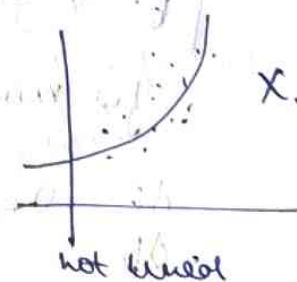


Relationship b/w

IP and O/P columns

should be linear.

No



not linear

2. Multi collinearity: There should be no multicollinearity (relation b/w IP cols)

$X_1 \quad X_2 \quad X_3 \quad | \quad Y$

No correlation

(should be independent)